BTECHBUDDY - Operating Systems Revision Notes

Process Synchronization, Scheduling & Deadlocks

Subject: CS204 | Operating Systems | 4th Semester

1. CPU Scheduling: FCFS, SJF Preemptive/Non-preemptive, Round Robin with Gantt charts.
2. Critical Section Problem: Mutual Exclusion, Progress, Bounded Waiting conditions.
3. Semaphores: Wait P and Signal V atomic primitives, Producer-Consumer problem.
4. Deadlock 4 Coffman Conditions: Mutual Exclusion, Hold & Wait, No Preemption, Circular Wait.
5. Bankers Algorithm: Safety algorithm and Resource Request algorithm example with Need matrix.
6. Memory Management: Paging, Page tables, TLB hit ratio calculation, Page Replacement FIFO, LRU, Optimal

BTECHBUDDY High-Yield Engineering Notes.